

Overview of Testing Techniques

Black Box Testing(Functional)

In black-box, or functional testing, test conditions are developed on the basis of the program or system's functionality; that is, the tester requires information about the input data and observed output, but does not know how the program or system works

The tester focuses on testing the program's functionality against the specification with black-box testing, the tester views the program as a black box and is completely unconcerned with the internal structure of the program or system.

Some examples in this category include decision tables, equivalence partitioning, range testing, boundary value testing, database integrity testing, cause–effect graphing, orthogonal array testing

White-Box Testing (Structural)

In white-box, or structural testing, test conditions are designed by examining paths of logic.

The tester examines the internal structure of the program or system.

Specific examples in this category include basis path analysis, statement coverage, branch coverage, condition coverage, and branch/condition coverage.

Gray-Box Testing (Functional and Structural)

Gray-box testing is a combination of black- and white-box testing.

The tester studies the requirements specifications and communicates with the developer to understand the internal structure of the system.

The motivation is to clear up ambiguous specifications and “read between the lines” to design implied tests. One example of the use of gray-box testing is when it appears to the tester that a certain functionality seems to be reused throughout an application.

Manual versus Automated Testing

The basis of the manual testing categorization is that it is not typically carried out by people and it is not implemented on the computer. Examples include structured walkthroughs, inspections, JADs, and desk checking.

The basis of the automated testing categorization is that it is implemented on the computer. Examples include boundary value testing, branch coverage testing, prototyping, and syntax testing. Syntax testing is performed by a language compiler, and the compiler is a program that executes on a computer.

Static versus Dynamic Testing

Static testing approaches are time independent and are classified in this way because they do not necessarily involve either manual or automated execution of the product being tested.

Examples include syntax checking, structured walkthroughs, and inspections. An inspection of a program occurs against a source code listing in which each code line is read line by line

Dynamic testing techniques are time dependent and involve executing a specific sequence of instructions on paper or by the computer. Examples include structured walkthroughs, in which the program logic is simulated by walking through the code and verbally describing it. Boundary testing is a dynamic testing technique that requires the execution of test cases on the computer with a specific focus on the boundary values associated with the inputs or outputs of the program.